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NMAM INSTITUTE OF TECHNOLOGY, NITTE
Off-Campus Centre of Nitte (Deemed to be University)
Sixth Semester B.Tech (CBCS) Degree Examinations

May 2025

EC4301-1 – MULTIMEDIA COMMUNICATION

Duration: 3 Hours

Max. Marks: 100

Note:

Part – A: Multiple Choice Questions: Answer all **Twenty** questions in the **OMR Sheet** provided. Each question carries equal marks.

Part – B: Descriptive Answer Questions: Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II each** and **One full** question from **Unit – III**.

PART - A: MULTIPLE CHOICE QUESTIONS**20 Marks**

1. The internet is made up of vast collection of interconnected networks all of which operate using the same set of ----- protocols.
A) Network
B) Communication
C) System
D) Service
2. Basic Rate Access supports
A) Two 32kbps channel
B) Two 64kbps channel
C) One 64kbps channel
D) One 32 kbps channel
3. In the world wide web, the standard format used for writing documents is known as -----
A) Marking Language
B) Text Language
C) Java script
D) Hypertext Markup Language
4. Which characteristic of multimedia streaming makes it different from traditional file downloading
A) Data is received in sequential order
B) It uses only lossless compression techniques
C) Playback begins before the entire file is downloaded
D) It always relies on TCP for reliability
5. What is the primary function of Real-time Transport Control Protocol (RTCP)?
A) Handle session establishment and call setup
B) Provide feedback on multimedia transmission quality
C) Transmit real-time media streams
D) Control congestion in TCP connections
6. In a multimedia communication system, which factor contributes most to packet loss?
A) Network Congestion
B) Packet size
C) Use of TCP instead of UDP
D) Code Selection
7. The total delay over a packet-switched network is the sum of -----
A) Forward delay, packet delay & Jitter
B) Forward delay, packetization delay & Jitter
C) Packetization delay, Mean network packet delay & Buffering Delay
D) Packet delay, Mean packet delay & Buffering Delay
8. The bit rate offered by G.728 standard is
A) 16kbps
B) 8kbps
C) 5.3kbps
D) 6.3kbps
9. The three features which determine the perception of a signal by the ear are its
A) pitch, amplitude and period
B) pitch, period and loudness
C) peak, period and loudness
D) peak, amplitude and pitch
10. What is the primary advantage of using RTP over raw UDP for multimedia communication
A) Guaranteed packet delivery
B) Support for encryption and authentication
C) Synchronization and real-time data transport
D) Reduced overhead compared to UDP

11. which mechanism helps to manage bandwidth allocation and prioritize real-time traffic
 A) Token Passing B) Circuit Switching
 C) Quality of Service (QoS) D) Carrier Sense Multiple Access (CSMA)
12. Letters m,v and l are examples for
 A) Pitch B) Period
 C) Voiced sounds D) Unvoiced Sounds
13. The accuracy of the prediction operation is determined by how well any movement between successive frames is estimated by an operation known as
 A) Motion estimation B) Motion compensation
 C) Motion correction D) Motion prediction
14. The effective rate of transmission of information bits is called the
 A) Data Rate B) Network throughput
 C) Bandwidth D) Bit Rate
15. Classic Ethernet is the original form of Ethernet that provides data rates between 3 to 10 Mbps
 A) 2 to 5 Mbps B) 3 to 10Mbps
 C) 2 to 20 Mbps D) 3 to 15Mbps
16. Token Ring is based on ----- token passing MAC protocol
 A) IEEE 802.1 B) IEEE 802.3
 C) IEEE 802.2 D) IEEE 802.5
17. ----- is a useful and inexpensive Internet based desktop videoconferencing system which allows users to communicate each other using audio and video technology
 A) CU-SeeMe B) EU-SeeMe
 C) SU-SeeMe D) MU-SeeMe
18. ----- is used to synchronize the two media streams prior to carrying out the decoding operation.
 A) Real-time Control Protocol B) Real-time Transport Control Protocol
 C) Real-time Transport Protocol D) Real-time Transmit Protocol
19. The four logical entities of H.323 standard are
 A) Base-stations, Gateways, Gatekeepers and Multipoint control units B) Terminals, Gateways, Routers and Multipoint control units
 C) Terminals, Gateways, Gatekeepers and Multipoint control units D) Base-stations, Gateways, Routers and Multipoint control units
20. The ----- identifier identifies the source device that has produced the packet contents.
 A) Contribution Source B) Token Source
 C) Synchronization Source D) Gateway Source

PART - B: DESCRIPTIVE ANSWER QUESTIONS

Unit – I		Marks	BT*	CO*	PO*
1.	a) Explain the following multimedia networks that are used to provide multimedia communication services: i. Broadcast television networks. ii. Broadband multiservice networks.	8	L*3	1	1
	b) With the aid of a neat diagram, illustrate the working of computer supported cooperative working.	8	L3	1	1
2.	a) A packet switched network with a worst-case jitter of 10ms is to be used for a number of applications each of which involve a constant bit rate information stream. Determine the minimum amount of memory that is required at the destination and a suitable packet size for each of the following input bit rates. It can be assumed that the mean packet transfer rate of the network exceeds the equivalent input bit rate in each case. i. 64 kbps ii. 256 kbps iii. 1.5 Mbps.	6	L3	1	1

- b) Explain the QoS parameters associated with:
- Packet-switched network.
 - Circuit-switched network.
- 10 L3 1 1
3. a) Explain with a neat diagram, the interactive applications for both cable and satellite network. 8 L3 2 1
- b) With the help of a diagram, describe the main components of a PSTN and show how a high-speed modem provides multiple services in addition to basic telephony. 8 L3 2 1

Unit – II

4. a) With the aid of block diagram, explain DPCM signal encoding and decoding principle. 8 L3 3 1
- b) Explain the implementation schematic and FIFO buffer operation for H.261 video encoder. 8 L2 3 1
5. a) A digitized video is to be compressed using the MPEG-1 standard. Assuming a frame sequence of: IBBPBBPBBPBBI... and average compression ratios of 10:1 (I), 20:1 (P) and 50:1 (B), derive the average bit rate that is generated by the encoder for both the NTSC and PAL digitization formats. 8 L3 3 1
- b) Explain the working principles of encoder and decoder for MPEG-4 standard. 8 L3 3 1
6. a) Discuss the following network performance parameters:
- Throughput
 - Networking delay
 - Delay variance
 - Error rate.
- 8 L3 4 1
- b) Explain the different modes of ATM employed in LANs to support multimedia traffic. 8 L3 4 1

Unit – III

7. a) Explain the usage of Real-time transport protocol. 8 L3 5 1
- b) Briefly discuss Real-time transport control protocol. 8 L3 5 1
8. a) Explain the system architecture and basic call flow associated with session initiation protocol. 8 L3 5 1
- b) Explain the architecture of H.323 standard. 8 L3 5 1

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
